

PATEL VRAJ GAURANGKUMAR

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PROFESSIONAL SUMMARY

B.Tech Information Technology student (CGPA 8.61/10) with hands-on experience across machine learning, computer vision, and full-stack web development. Has independently built and deployed five end-to-end projects spanning AI-powered image restoration, driver monitoring, financial analytics, civic-tech, and attendance automation. Looking for a Data Science, AI/ML, or Software Development role to keep building things that solve real problems.

EDUCATION

B.Tech in Information Technology

Expected May 2027

Madhuben & Bhanubhai Patel Institute of Technology (MBIT), Anand, Gujarat

Cumulative CGPA: 8.61/10.00 | SGPA: S1 8.83, S2 8.17, S3 8.48, S4 8.30, S5 8.75, S6 9.19

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML5, CSS3, JavaScript (Basic)

ML / Python Libraries: PyTorch, TensorFlow, OpenCV, Scikit-learn, MediaPipe, NumPy, Pandas, Matplotlib, Streamlit

Frameworks & Tools: Flask, Gradio, Git, GitHub, SQLite, PostgreSQL, YOLOv11

PROJECTS

NeuralRestore — AI-Powered Image Restoration

2026

Python, PyTorch, Gradio, OpenCV

- Built a sequential 5-model deep learning pipeline that automatically detects and repairs damage in old or corrupted images with no manual masking, combining EfficientNet-B4 + UNet for damage/mask detection, LaMa (FFC) for inpainting, DnCNN for denoising, NAFNet for deblurring, and Real-ESRGAN 4x for super-resolution.
- Added test-time augmentation (4 flip variants) and configurable mask thresholds to improve restoration consistency; shipped a drag-and-drop Gradio UI with before/after comparison that runs fully offline with zero cloud dependency.

Driver Drowsiness & Distraction Detection System

March 2026

Python, MediaPipe, YOLOv11, OpenCV

- Built a real-time dual-phase system using MediaPipe Face Mesh for EAR-based drowsiness detection and head pose estimation, with auto-calibration and 8-frame smoothing to cut false positives.
- Trained a custom YOLOv11 model on 2,004 images across 2 classes to detect phone usage, and integrated independent drowsy/distraction/phone alarms with CLAHE + gamma correction for low-light performance.

StockSense AI — AI-Powered Stock Analysis Chatbot

2025

Python, Flask, XGBoost, GARCH, Plotly, yfinance

- Built a WhatsApp-style chatbot for natural-language stock queries on any global ticker, with interactive Plotly charts for RSI, MACD, Bollinger Bands, and SMA.
- Implemented Buy/Sell recommendations using a hybrid RSI + MACD + SMA + GARCH + XGBoost model, with multi-stock comparison and a financial concept explainer.

Anand Civic Issue Reporting System

2025

Python, Flask, JWT, SQLite/PostgreSQL, Leaflet.js

- Built a full-stack, role-based (Citizen, Officer, Dept. Admin, Municipal Admin) complaint platform with GPS-based Anand city boundary validation, image uploads, and optional Gemini AI image analysis.
- Designed a 6-stage complaint workflow with JWT auth, bcrypt hashing, rate limiting, automated notifications, and Chart.js analytics.

Smart Attendance System

2024

Python, Flask, Face Recognition, SQLite

- Built a web-based attendance system with face recognition, real-time tracking, and role-based dashboards for Admin, Teacher, and Student, with CSV/PDF report export.
- Implemented multi-role authentication and a Unicorn deployment pipeline, including a custom build script for compiling the C++ face-recognition library.

ACTIVITIES & HACKATHONS

CVMU Hackathon 4.0 (36-Hour)

2026

- Presented the Anand Civic Issue Reporting System project at a 36-hour university-level hackathon hosted by Charotar Vidya Mandal University, delivering a fully working prototype under strict deadlines through rapid prototyping and agile teamwork.

Internal College Hackathon — SIH Selection Round

2025

- Presented the Smart Attendance System project in MBIT's internal Smart India Hackathon selection round, collaborating in a team to ideate, build, and present a socially impactful technical solution.